

PERIOSTEAL BONE REACTION

PERIOSTEUM

- Fibrous connective tissue membrane that covers the external surfaces of all bones with the exception of joint surfaces, which are covered by articular cartilage.
- The periosteum has thick collagen fibers called Sharpey's fibres which penetrate perpendicularly into the outer cortex, anchoring the periosteum to the underlying bone.
- Two layers: Outer fibrous layer & Inner fibrous layer

LAYERS OF PERIOSTEUM

- Outer fibrous layer: Subdivided into deep and superficial - more collagenous matrix, less cells - fibroblasts.

Superficial - rich in blood vessels, nerves, and few elastic fibres.

Deep - many elastic fibres , few blood vessels.

- Inner cellular layer/ Cambium layer: Less collagenous matrix, more cells, osteogenic progenitor cells, osteoblasts.

FUNCTIONS OF PERIOSTEUM

- Nourishment to outer cortex of bone.
- Pain sensation
- New bone formation, bone remodeling, fracture healing

Periosteal bone reaction

- Formation of new bone in response to abnormal stimuli.
- Stimuli - infection, tumour , arthritic conditions, any other inflammatory irritant.
- These cause subperiosteal extensions of blood, pus, or tumour . These migrate beneath the periosteum by way of the Haversian canals, and cause elevation of periosteum.
- The appearance of periosteal reaction is determined by the intensity, aggressiveness, and duration of the underlying insult.
- The periosteum in children is more active and less adherent to the cortex than in adults. Thus, periosteal reaction can occur earlier and appear more aggressive in children than in adults.

TYPES OF PERIOSTEAL REACTION

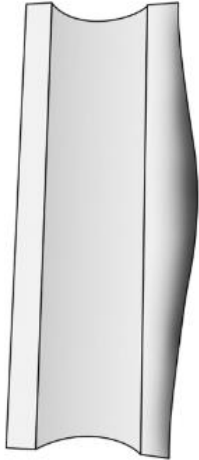
NONAGGRESSIVE

- Thin
- Solid
- Thick irregular
- Septated

AGGRESSIVE

- Laminated (Onion skin)
- Spiculated - Perpendicular/Hair-on-end
 - Sunburst
- Disorganized
- Codman triangle

THIN TYPE



- A single lamella is one sheet of new bone a millimeter or two away from the cortical surface. It may or may not join cortex at its extremes.
- The single lamella reaction has been called the hallmark of a benign process.
- Acute osteomyelitis, eosinophilic granuloma, storage disease.

SOLID TYPE



- In this pattern, the reaction is localized and caused by a focal insult. Usually, its form is elliptical and dense, resulting in cortical thickening.

- It happens because of successive and gradual addition of compact bone to the surface, until merging of layers.

- Causes: - Osteoid osteoma

- Osteomyelitis

- Osteosarcoma

- Osteoblastoma

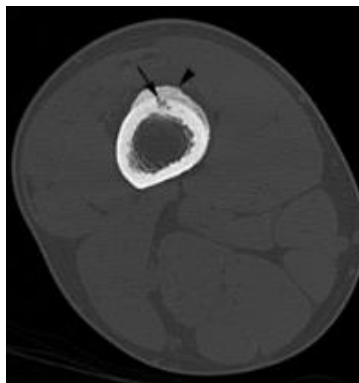
- Chondrosarcoma

- Chondroblastoma

- Ewing sarcoma

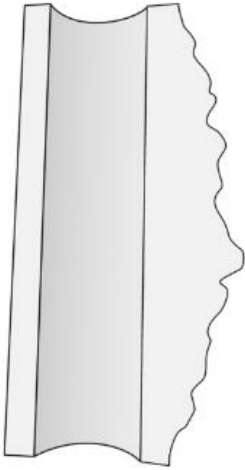


Solid periosteal reaction (osteoid osteoma). Lateral radiograph reveals thick, smooth periosteal reaction in anterior cortex of distal femur (arrow).



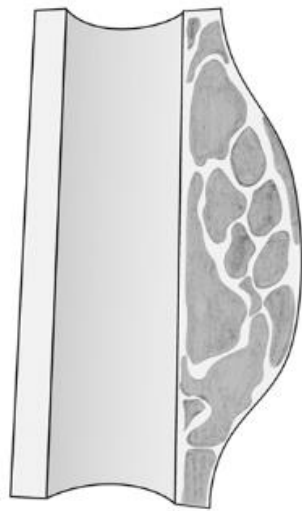
Solid periosteal reaction (osteoid osteoma). Axial CT image shows lucent central nidus (arrow) of osteoid osteoma and thick reactive periosteal reaction (arrowhead).

THICK IRREGULAR TYPE



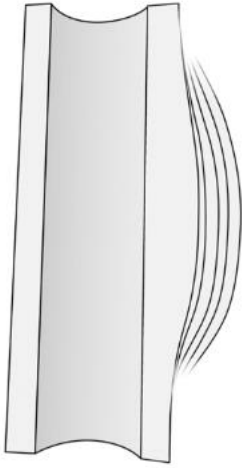
- The presence of the thick irregular pattern of periosteal reaction may imply a more temporally and spatially heterogeneous process such as Chronic osteomyelitis, Venous stasis.

SEPTATED



- A type of non-aggressive , layered bone formation where a periosteal reaction is divided by internal septations, or thin internal walls.
- Conditions: Fibrous dysplasia
 - Aneurysmal bone cyst
 - Giant cell tumour

MULTIPLE LAMELLAR (ONION SKIN)

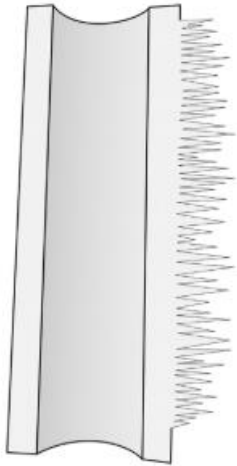


- It is caused by repeated periosteal stimulation without time for incorporation into the cortex.
- On plain films, it appears as multiple continuous lamellae with lucent zones between them, widening the bone contour.
- Lamellated reactions indicate a rapidly progressive lesion and are usually seen with bone sarcomas, especially Ewings sarcoma and osteosarcoma, acute osteomyelitis, occasionally stress fractures.

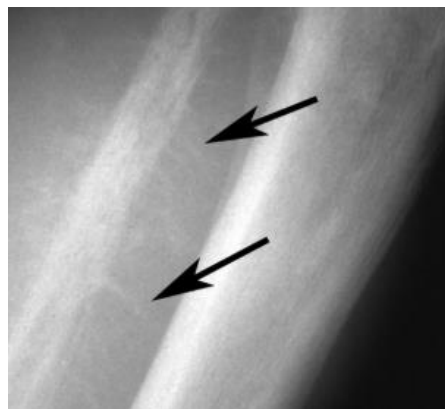


Onionskin periosteal reaction (osteomyelitis). Frontal radiograph shows localized laminated periosteal reaction (arrow) along lateral cortex of distal femur.

HAIR ON END / PERPENDICULAR



- Denotes a more rapid underlying process.
- The spicules are higher in the midzone of the reaction, decreasing in height proximally and distally.
- Its configuration is actually an interconnected compartmental system resembling a honeycomb. Compartment walls lying parallel to the radiographic beam register as spicules, whereas walls crossed perpendicularly by the beam are inapparent.
- This pattern is seen in malignant tumours such as Ewings sarcoma, osteosarcoma.
- Hyperplastic marrow in thalassemia.
- This pattern favors a malignant tumor but may be stimulated by certain inflammatory states like syphilis, myositis, and Caffey's disease

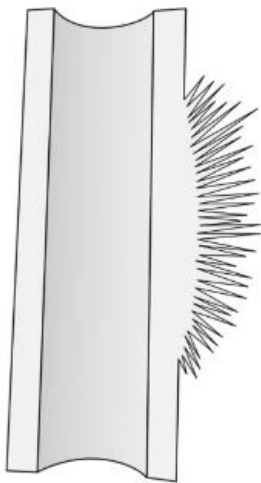


Hair-on-end periosteal reaction (Ewing's sarcoma). Lateral radiograph of lower leg shows bony spicules emanating perpendicular to cortex (arrows).



Hair-on-end periosteal reaction (chronic osteomyelitis). Axial CT image of shoulder shows spiculations (arrows) arising along the posterior cortex of scapula.

SUNBURST / DIVERGENT SPICULATED

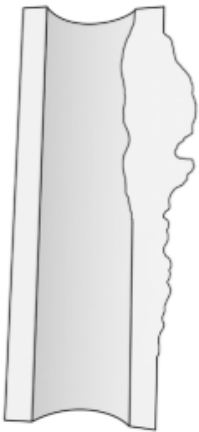


- Denotes malignant osteoid production and only partially consists of reactive bone. It consists of streaks of variable thickness and orientation converging on a medullary epicentre.
- The rays are reactive bone, and the space between rays in osteosarcoma is filled with neoplastic cells and tumor products, usually chondroid and myxoid.
- Highly suggestive of osteosarcoma
- May also be seen with blastic metastasis and osseous hemangioma, especially in the skull.



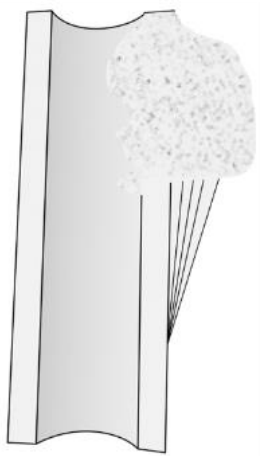
Sunburst periosteal reaction (osteogenic sarcoma complicating long-standing Paget's disease). Frogleg radiograph of femur shows sunburst and disorganized aggressive periosteal reaction (arrows).

DISORGANIZED



- Appearance: Spicules of new bone are disorganized and have a random orientation.
- Conditions: Associated with malignant bone tumors like osteosarcoma or Ewing sarcoma.

CODMANS TRIANGLE



- The Codman triangle indicates a periosteal elevation caused by mass effect.

- A Codman triangle is a cuff of reactive bone surrounding the upper and/or lower limits of a lesion on one or both sides of the shaft.
- Radiographically, a triangle is formed at the transition between the normal periosteum and the adjacent elevated periosteum.
- It has a triangular shape, caused by a space between the neoplastic mass or blood, for example, and the periosteal thin line.
- Usually seen with aggressive malignant tumors breaking out into the soft tissue, such as osteosarcoma and chondrosarcoma, but sometimes occurs as a defense against inflammation in acute osteomyelitis.



Codman triangle (prostate cancer). Frontal radiograph of distal femur shows the edge of periosteum (thin arrow) lifted off cortex (arrowhead) at site of sclerotic metastasis (thick arrow).

PERIOSTEAL REACTION CAUSES:

- Arthritis
 - Psoriatic arthritis
 - Reactive arthritis
- Metabolic
 - Hypertrophic pulmonary osteoarthropathy
 - Thyroid acropathy
- Congenital
 - Pachydermoperiostosis
 - periosteal reaction of newborn
- Trauma
 - Stress fracture
 - Fracture
- Drugs
 - Fluorosis

- Hypervitaminosis A
- Prostaglandins
- Tumors
 - Osteosarcoma
 - Ewings sarcoma
 - Chondroblastoma
 - Eosinophilic granuloma
 - Osteoid osteoma
 - Leukemia and lymphoma
- Infection
- Genetic
 - Caffey disease
- Vascular
 - Venous stasis

OSTEOSARCOMA

- Conventional osteosarcomas are common high-grade intramedullary neoplasms that produce an osteoid matrix.
- The majority of lesions occur in patients under 25 years old, with the femur, tibia, and fibula the most common sites. The sunburst, hair-on- end, or Codman triangle subtypes of periosteal reaction are most frequently seen.
- However, laminated, solid, thin, or disorganized forms of periosteal reaction can also be present.
- A wide zone of transition, cortical breakthrough, and soft-tissue mass are all concerning features that warrant further evaluation.



Osteogenic sarcoma. Lateral radiograph of mid femur shows sunburst periosteal reaction with bone formation in divergent pattern (arrow).

EWINGS SARCOMA

- It is derived from undifferentiated mesenchymal cells of bone marrow or primitive neuroectodermal cells and accounts for 6–8% of primary malignant bone tumors.
- Although characteristically intramedullary in location, on radiographs, only the cortical changes may be apparent with a permeative or moth-eaten osteolytic component.
- A large soft-tissue mass can be seen. The periosteal reaction ONION PEEL (MULTIPLE LAMELLETED) pattern is typically aggressive, with the hair-on-end subtype highly characteristic for Ewing's sarcoma.

CHONDROBLASTOMA

- Chondroblastomas are benign cartilage-producing lesions that typically occur in the epiphyses of skeletally immature patients.
- The lesions are typically lytic and may have a sclerotic margin.
- Periosteal reaction due to chondroblastoma most commonly occurs in large lesions in flat or small tubular bones.
- The periosteal reaction can be thick, solid, or laminated.



Chondroblastoma. Radiograph shows laminated (arrow) and disorganized periosteal reaction along proximal humerus.

OSTEOID OSTEOMA

- It is a benign bone-forming tumor affecting children and adolescents, most commonly occurring in the femur, tibia, fibula, or humerus.
- A thick and dense periosteal reaction develops as a response to the tumor.
- The central lucent nidus may be difficult to visualize on radiographs, and CT can be helpful in these instances.

- Subperiosteal osteoid osteomas can produce extensive aggressive periosteal reaction, whereas intraarticular lesions typically cause relatively little periosteal new bone formation.



Osteoid osteoma. Frontal radiograph of proximal tibia shows smooth, thick periosteal reaction along medial tibia cortex (arrow).

CONCLUSION

Periosteal reaction results from the response of cortical bone to a variety of insults. Recognition of the presence of periosteal reaction is the most important step. Occasionally, the pattern of periosteal reaction is highly suggestive of a particular process, but in general there is significant overlap in the disease entities that result in aggressive and nonaggressive forms. Intense, rapid-acting processes usually result in aggressive periosteal reaction; slower, indolent processes result in a nonaggressive form. The causes of periosteal reaction are broad, including trauma, infection, arthritis, tumors, and drug-induced and vascular entities. When periosteal reaction occurs in a bilateral distribution, a systemic disease process should be considered.

REFERENCE

- White & Pharoah's oral radiology: principles and interpretation
- Shafer's Textbook of oral pathology